# 020116 4D56 Hi-Power Overheat –Working procedure (revision 3)

**Cautions:**

Updated 09/10/2014 (Page 7)

Update 01/06/2016 (Page 2 and 3) Step 0

1) Use the supplied “Warranty Application” to produce the technician working sheet for each vehicle. This is to ensure the correct combination “upgrades” are performed for the specific vehicle.

   A “sample” technician working sheet is shown on page 2 of this document.

2) Follow each step in the correct order **EXACTLY** as nominated in the detailed working procedure.

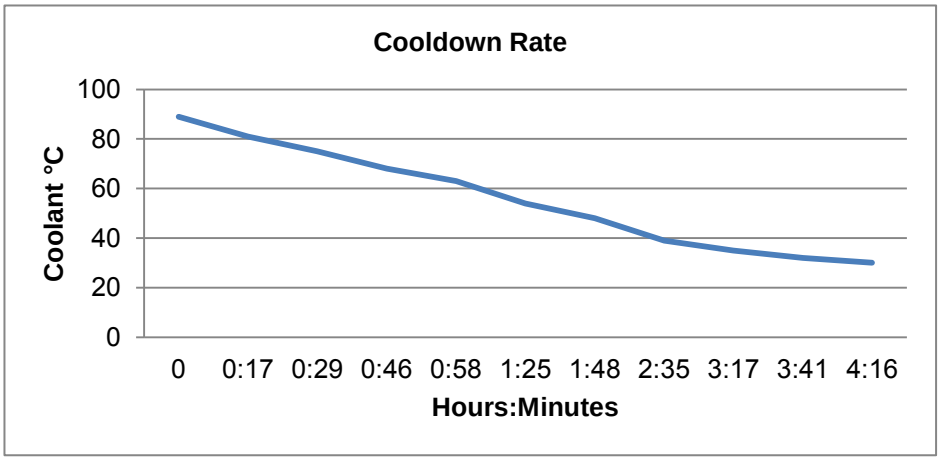
3) Record the results and clock times at each step.

4) If the steps are not followed correctly, this may result in unnecessary repairs being performed.

5) Failure to document the step results and the clock time where required, may result in any repair claim being overturned during the audit process.

6) Note: If you need to perform Step 5 (Engine Coolant Blow-by Inspection) ensure the coolant is drained and refilled with 50% concentration with the **air bleeding performed as per the service manual** (Item 14 ENGINE COOLING - ON-VEHICLE SERVICE – ENGINE COOLANT REPLACEMENT).

   Note: Ensure you allow sufficient time for the engine cool-down periods (X 2) as per the chart below.




## 1. Necessary parts

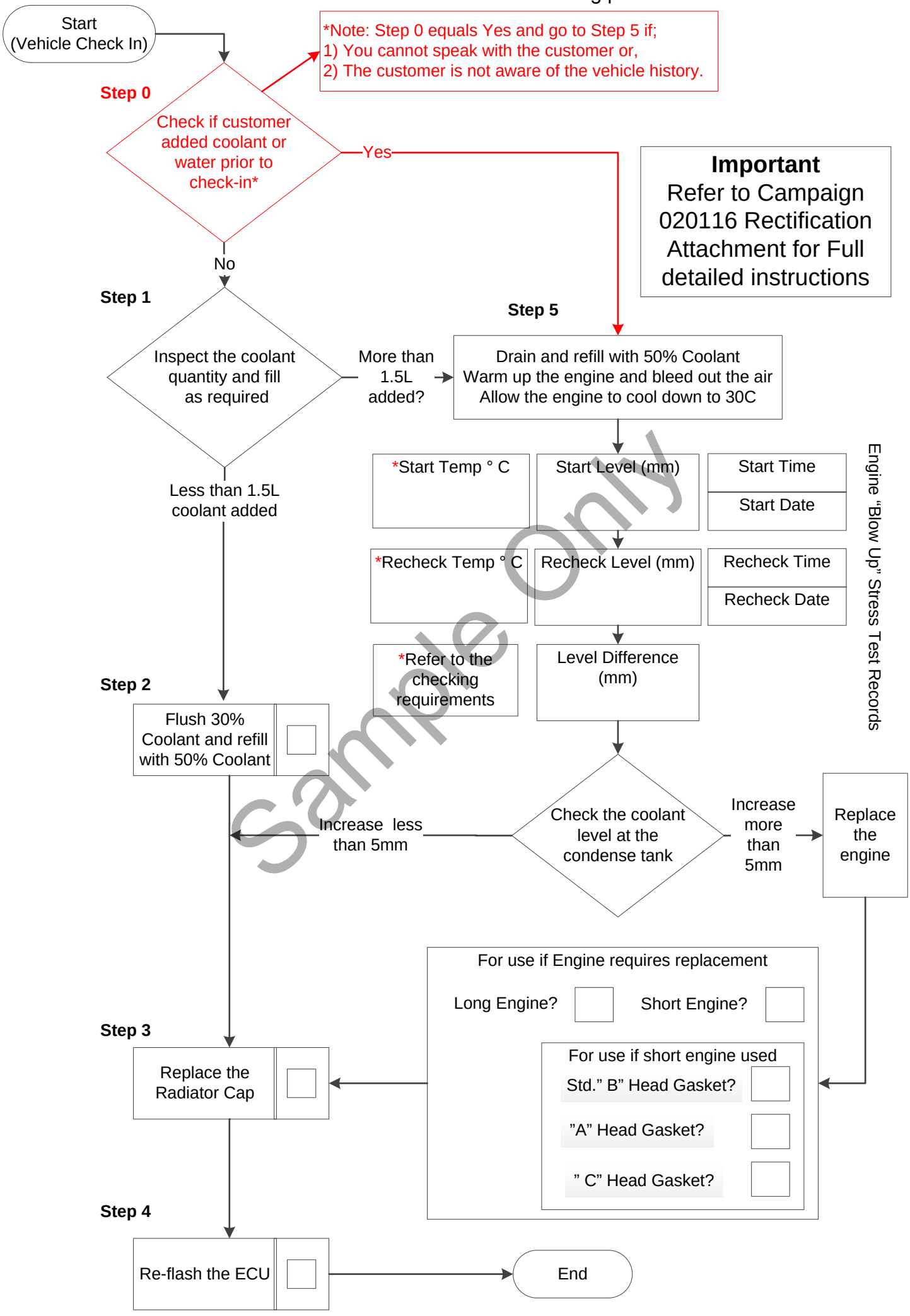
1) Part to be replaced (this part is used by all vehicles in this campaign)

| Part Name | Part No. | Required Qty./vehicle | Remarks |
|---|---|---|---|
| CAP, RADIATOR | FOC-RADCAP | 1 | Valve opening pressure: 127kPa |

2) In case “Engine Coolant Internal Leakage” between cylinder bores is confirmed, replace the engine referring to the allowed parts listed on pages 4 or 5 of the 020116 Dealer Notification document.

   Note: It is recommended to use the newly introduced “long engine” when “Engine Coolant Internal Leakage” is identified.
   Only use the short engine method if a long engine is not available.

   Note: ENSURE you follow the inspection procedure EXACTLY documenting each step and result on the forms generated from the provided “Warranty Application.


Start
(Vehicle Check In)
*Note: Step 0 equals Yes and go to Step 5 if;
1) You cannot speak with the customer or,
2) The customer is not aware of the vehicle history.
Step 0
Check if customer added coolant or water prior to check-in*
Yes
No
Important
Refer to Campaign 020116 Rectification Attachment for Full detailed instructions
Step 1
Inspect the coolant quantity and fill as required
More than 1.5L added?
Step 5
Drain and refill with 50% Coolant
Warm up the engine and bleed out the air
Allow the engine to cool down to 30C
Less than 1.5L coolant added
*Start Temp ° C
Start Level (mm)
Start Time
Start Date
*Recheck Temp ° C
Recheck Level (mm)
Recheck Time
Recheck Date
*Refer to the checking requirements
Level Difference (mm)
Engine "Blow Up" Stress Test Records
Step 2
Flush 30% Coolant and refill with 50% Coolant
Check the coolant level at the condense tank
Increase less than 5mm
Increase more than 5mm
Replace the engine
For use if Engine requires replacement
Long Engine?
Short Engine?
For use if short engine used
Std." B" Head Gasket?
"A" Head Gasket?
" C" Head Gasket?
Step 3
Replace the Radiator Cap
Step 4
Re-flash the ECU
End

Sample Only

## 2. Work procedure

**[Step 0] Check if the customer has added coolant or water to the system.***

*__Note:__ If you cannot speak with the customer or, the customer is not aware of the vehicle history or overheating has been reported, go to (Step 5).

a) In case coolant NOT added and overheat was NOT reported, go to (Step 1)

b) In case coolant WAS added or overheat WAS reported, go to (Step 5)

**[Step 1] Inspection of the coolant level**

- Remove the current radiator cap and fill up the coolant (or water*) in the radiator.

  * In case concentration of coolant in the system is known to be 30% (Coolant Change listed under the "Defined Work Operations for this Vehicle"), tap water can be used since all coolant will eventually be drained out and the 50% coolant will be used to refill the system.

  In case that the concentration of coolant system is 50% (Coolant Change **NOT** listed under the "Defined Work Operations for this Vehicle"), 50% coolant must be used for this check.

- Install the current radiator cap back on and then carry out air bleeding in accordance with the procedure shown in the Workshop Manual (Item 14 ENGINE COOLING - ON-VEHICLE SERVICE – ENGINE COOLANT REPLACEMENT).

**13.** Start the engine and let it warm up until the thermostat opens.
**14.** After repeatedly revving the engine up to 3,000 r/min several times, then stop the engine.
**15.** Remove the radiator cap after the engine has become cold, and pour in engine coolant up to the brim. Reinstall the radiator cap.
**16.**

| **caution** | Do not overfill the radiator condenser tank. |
|---|---|

Add engine coolant to the radiator condenser tank between the "F" and "L" mark if necessary.

- Measure the total quantity of coolant added to fill up the system (at the end of the air bleeding process).

  <In case the quantity of added coolant is less than 1.5L>

  Go to **[Step 2]**

  < In case the quantity of added coolant is 1.5L or more>

  Go to **[Step 5]**

**[Step 2] Coolant drain and refill**

- Drain all coolant and fill up the system with new coolant of 50% concentration.

  (8.2L of new coolant is required to fill up the system including reservoir tank.)

  * This work is required only for vehicles when listed on the "Working Procedure."

**[Step 3] Replacement of radiator cap**

- Replace radiator cap with the one whose valve opening pressure is 127kP. The valve opening pressure is printed on the cap as shown in the photo below..




Current P/No. MR597126
(Valve opening press : 108kPa)

New P/No. 1350A920
(Valve opening press : 127kPa)

Go to **[Step 4]** after replacing the radiator cap

**[Step 4] Re-programming engine ECU**

- Reprogram the engine ECU using the CD-ROM **CD3A14051**

  (sent to dealers as detailed in General Letter GL-14-019)

  **Note: Record the starting and final Software P/N for reference on the repair order**

Refer to the table below for the update software.

| Specification of the vehicle | Data No. | ECU P/N | Software P/N | |
|---|---|---|---|---|
| | | | New | Old |
| 10MY CR MMAL 4D56 EURO4 5AT 4WD H/P | 1862B775 | 1860B241 | 1860B241 06 | 1860B241 05 - 01 |
| 10MY CR MMAL 4D56 EURO4 5MT 4WD H/P | 1862B776 | 1860B347 | 1860B347 04 | 1860B347 03 - 01 |
| 10MY CR MMAL 4D56 EURO4 4AT 4WD H/P | 1862B777 | 1860B348 | 1860B348 07 | 1860B348 06 - 01 |
| 11-13MY CR MMAL4D56 EURO4 5MT 2/4WD H/P | 1862B778 | 1860B583 | 1860B583 06 | 1860B583 05 - 01 |
| 11-14MY CR MMAL 4D56 EURO4 4AT 2/4WD H/P | 1862B770 | 1860B584 | 1860B584 07 | 1860B584 06 - 01 |
| 11-14MY CR MMAL 4D56 EURO4 5AT 2/4WD H/P | 1862B771 | 1860B585 | 1860B585 06 | 1860B585 05 - 01 |
| 14MY CR MMAL 4D56 EURO4 5MT 2/4WD H/P | 1862B772 | 1860C267 | 1860C267 02 | 1860C267 01 |
| 09-10MY CR45 MMAL 4D56 EURO4 5AT 4WD H/P | 1862B779 | 1860B208 | 1860B208 05 | 1860B208 04 - 01 |
| 09-10MY CR45 MMAL 4D56 EURO4 5MT 4WD H/P | 1862B780 | 1860B210 | 1860B210 05 | 1860B210 04 - 01 |
| 11-13MY CR45 MMAL 4D56 EURO4 5MT 2/4WD H/P | 1862B781 | 1860B554 | 1860B554 06 | 1860B554 05 - 01 |
| 11-14MY CR45 MMAL 4D56 EURO4 5AT 2/4WD H/P | 1862B773 | 1860B555 | 1860B555 05 | 1860B555 04 - 01 |
| 14MY CR45 MMAL 4D56 EURO4 5MT 4WD H/P | 1862B774 | 1860C272 | 1860C272 02 | 1860C272 01 |

**<End of Work>**

**[Step 5] Internal Coolant Blow-by Inspection**

In case the quantity of added coolant is 1.5L or more in **[Step 1],** there is a possibility that Engine Coolant Blow-by may have been occurring.

**To confirm this, the following inspection is required.**

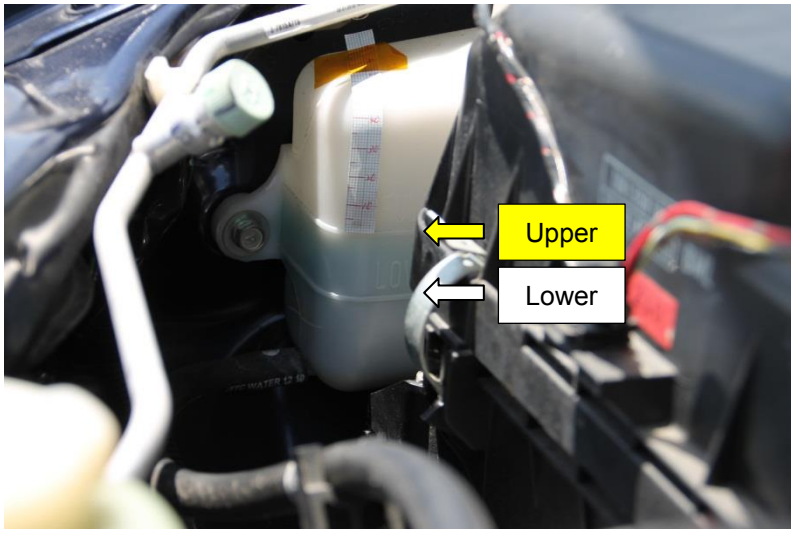
1) Use the original radiator cap (valve opening pressure 108kPa) for the inspection.
2) **For vehicles using 30% coolant (coolant change listed in the Working Procedure), drain and refill the system with 50% coolant first.**

   Add 50% coolant to the system; fill up radiator and also refill reservoir to the upper level **(this process applies to all vehicles).**
3) Carry out air bleeding procedure in accordance with the Workshop Manual (Item 14 ENGINE COOLING - ON-VEHICLE SERVICE – ENGINE COOLANT REPLACEMENT). Turn the engine off and check the coolant level at the radiator and reservoir tank again. Then refill coolant to the radiator and reservoir tank up to upper level.
4) Cool the coolant down to 30 degree Celsius or lower.

   If the ambient temperature is over 30 degree Celsius, cool the coolant down to the ambient temperature. (Please try to find a location where ambient temperature is below or close to 30 degrees).

   Note: Record the coolant temperature and the time of the check on the Working Procedure form.
5) Check the coolant level at the reservoir tank to see if it stays at the upper level.

   Note: In case that the level is below the upper level, refill the coolant to the upper level.




6) Close the engine hood and start the engine.

   Keep running for approximately 15 minutes under the following conditions:
   - Park brake applied
   - Shift position: P (A/T) or N (M/T)
   - Engine speed: 4,000 rpm constant.

7) Turn the engine off 15 minutes after the start-up.

Note: Cool the coolant down to the temperature recorded in Step 5 Item 4).

Check the coolant level at the reservoir tank.

Record the coolant temperature, level and time of the check on the Working Procedure form.

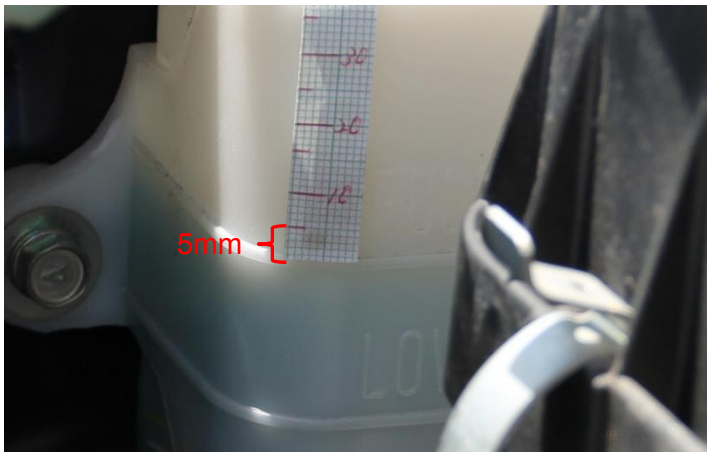
* Use electric fan to cool down the system if necessary.

<In case the coolant level is less than 5 mm above the upper level>

Go back to **[Step 3]** (no engine coolant blow-by between cylinder bores is occurring)

<In case the coolant level is higher than the upper level by 5 mm or more>

Go to **[Step 6]** (there is a possibility that engine coolant blow-by between cylinder bores is occurring)



**[Step 6] Engine Replacement**

If the vehicle fails the engine coolant blow-by inspection, the engine may be prone to experience internal coolant leakage and engine replacement is required. Replace with a Long Engine Assembly, or if not available, a short engine together with cylinder head.

**Note:** Initial stocks of long engines are limited so stocks are restricted. After placing the parts order for the required engine, send an email to customerrelations@mmal.com.au explaining the specific circumstances for the vehicle in question including a copy of the completed working procedure and the SAP order and engine part number that was ordered.
For the engine replacement procedure, refer to the relevant Workshop Manual.
Component parts for the long engines assemblies are shown in the table below for reference.

Note: Components not listed must be removed from the original engine and installed on the **replacement long engine** (after ensuring they are within the service limits) with the exception of the injection pipes shown below.
Note: As the Cam Sprocket and Timing belt tensioner adjuster have not been fitted to the new long engines, the valve clearances have NOT been set and the rocker arm lock nuts have been left loose. After fitting the (original) Cam Sprocket, (original) Timing belt tensioner adjuster, (original) Fuel pump and the (new) Timing belt, remove the valve cover, adjust the valve clearances and correctly tighten the rocker arm lock nuts.

| Component parts in the long engine |
|---|
| CYLINDER BLOCK |
| CRANKSHAFT & BEARING |
| PISTON |
| CONNECTING ROD & BEARING |
| TIMING BELT TRAIN (excluding Cam Sprocket and Timing belt tensioner adjuster) |
| BALANCER, BALANCER BELT TRAIN |
| CYLINDER HEAD & GASKET |
| VALVE GEAR TRAIN |
| ROCKER COVER |
| CAMSHAFT |
| OIL PAN |
| TIMING BELT COVER |
| FRONT CASE |
| VACUUM PUMP |
| WATER PUMP |
| OIL PUMP & GEAR TRAIN |
| OIL FILTER, OIL FILTER BRACKET |
| OIL STRAINER & PIPE |
| OIL PRESSURE SW OR GAUGE UNIT |
| OIL COOLER |
| HANGER,ENGINE |
| FUEL INJECTOR |
| FUEL LEAKAGE PIPE |
| CRANK ANGLE SENSOR |
| GLOW PLUG |

Note: The long engine is equipped with NEW fuel injectors, so these pipes MUST be replaced with NEW parts.

| Description | Part No. | Qty |
|---|---|---|
| TUBE-FUEL INJECTION-NO.1 | 1428A044 | 1 |
| TUBE-FUEL INJECTION-NO.2 | 1428A166 | 1 |
| TUBE-FUEL INJECTION-NO.3 | 1428A045 | 1 |
| TUBE-FUEL INJECTION-NO.4 | 1428A034 | 1 |
| TUBE SET-FUEL INJECTION | 1428A165 | 1 |

Go back to **[Step 3]** after the completion of the engine replacement.